

Session 2 · Visit by PT Passion Teknologi Inovasi (PLABS), ELSPINA VEINZ, and KUSHIDA KOGYO to FMIPA ITB

Thursday, 17 September 2026 · Physics Building, Ganesha Campus

Rapid Identification of Vegetable Oils with Multi-Wavelength NIR LEDs Assisted by Machine Learning

A low-cost LED–photodiode sensor combined with machine learning

Research and Developer Team:

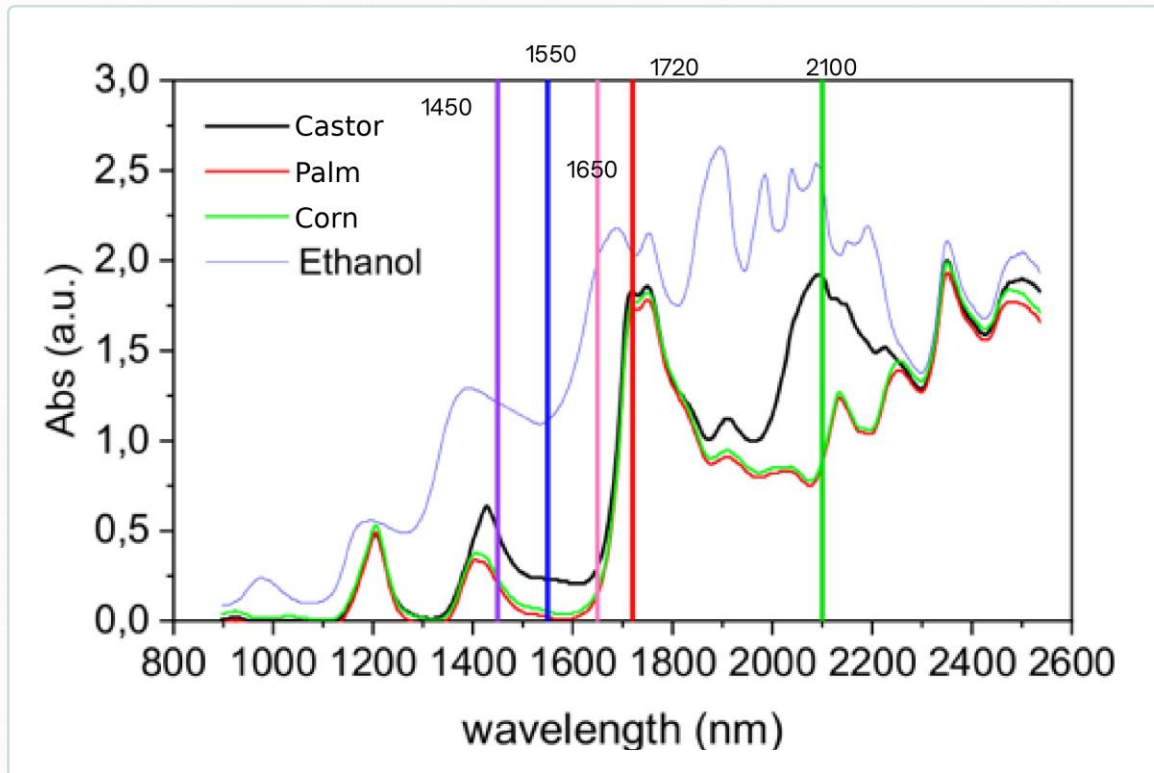
Prof. Rahmat Hidayat

Dr. Sparisoma Viridi

Aulia Defitri Wulandari, M.Si

Susi Komalasari, S.Si

The NIR fingerprint: small differences at particular bands



Full NIR absorbance spectrum (800–2600 nm); vertical lines = the five LED wavelengths.

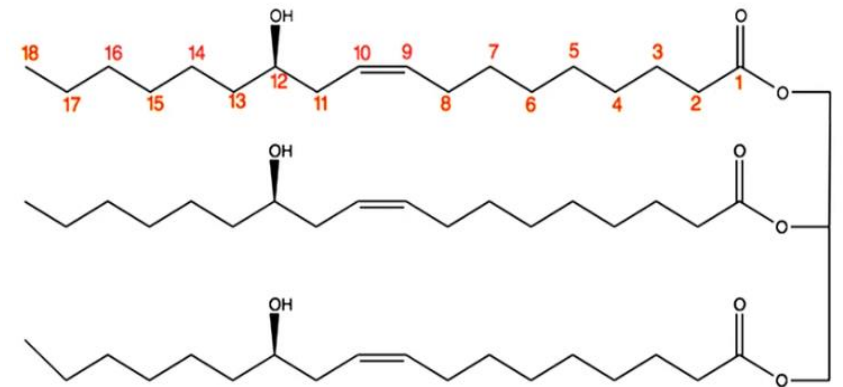
Key idea: choose the wavelengths that hold the most informative chemical contrast.

1 What NIR reads

C–H and O–H bonds absorb through overtone and combination transitions.

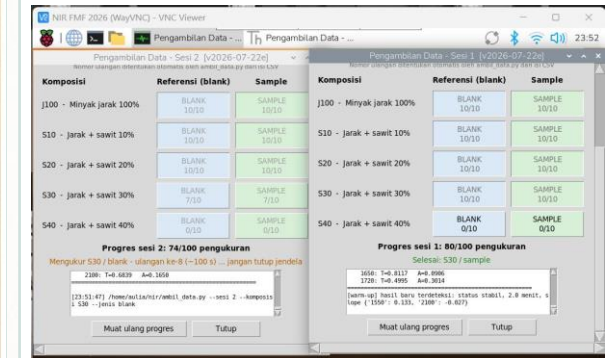
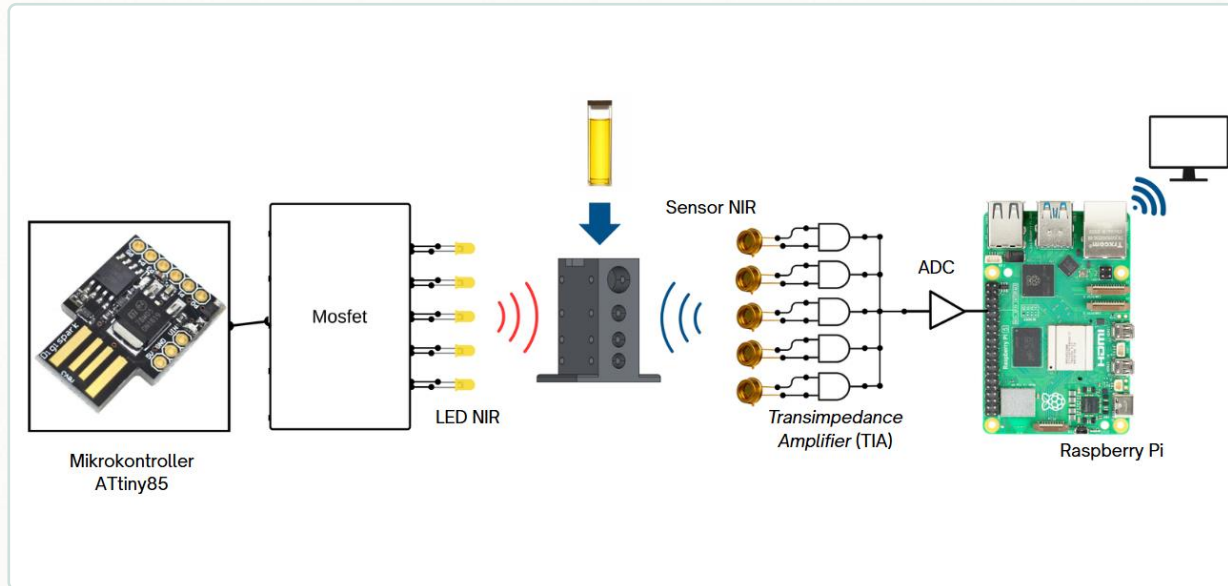
2 Why each oil differs

Chain length, degree of unsaturation, and hydroxyl groups shape a characteristic absorbance pattern.



Chemical structure of ricinoleic acid, the main component of castor oil

Portable NIR LED–Photodiode System



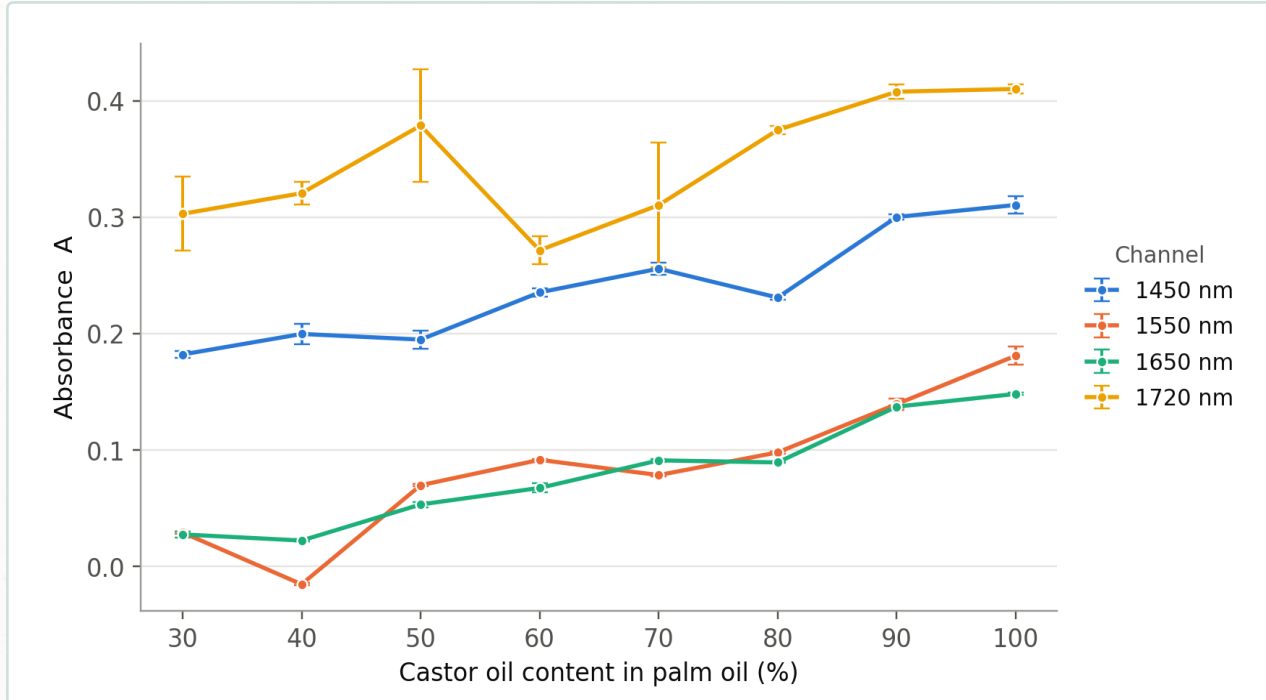
**ATtiny85
microcontroller**
replaces the waveform
generator and
oscilloscope

**Embedded TIA
amplifier**
amplifies the
photodiode signal on
one board

Raspberry Pi
replaces the
electrometer;
processes and displays
the result

**One 3D-printed
enclosure**
the whole system
integrated, ready for
the field

The sensor response follows the blend level

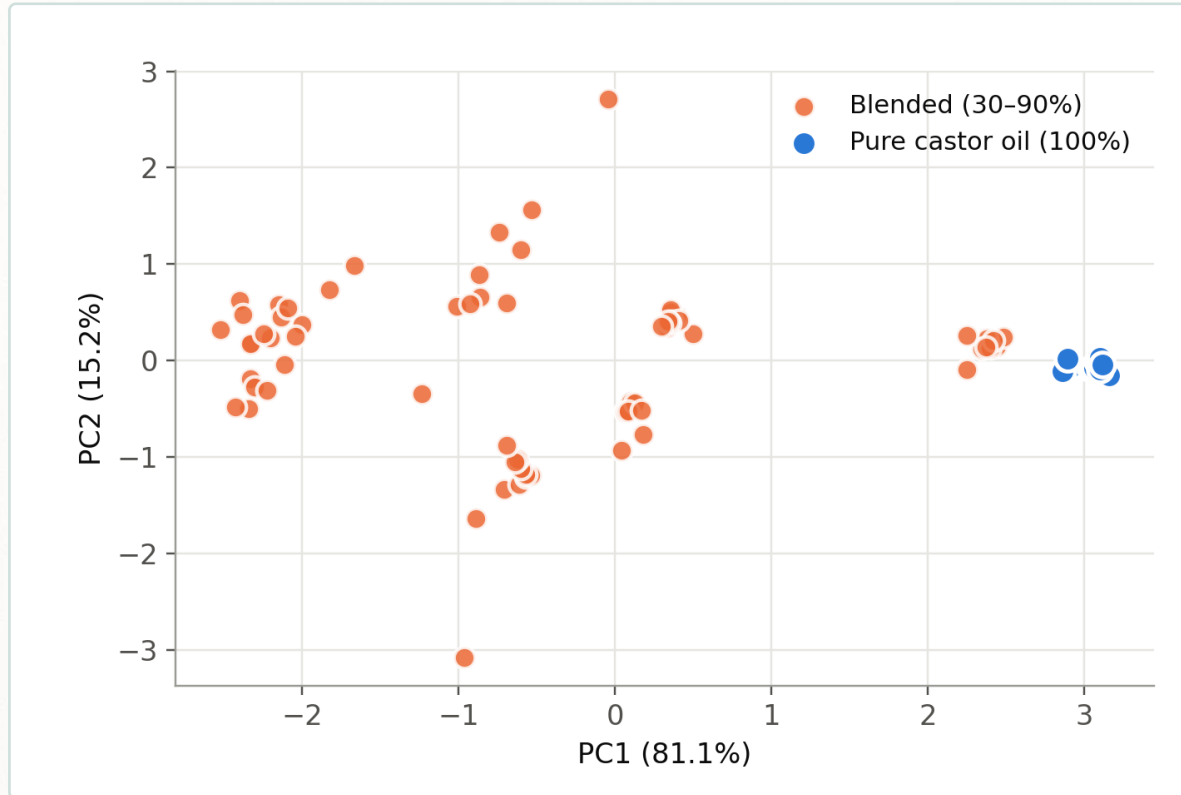


Mean absorbance against castor oil content in palm oil. Vertical bars = standard deviation of 10 repeats: 0.005 (1450 nm) · 0.002 (1550 nm) · 0.002 (1650 nm) · 0.021 (1720 nm).

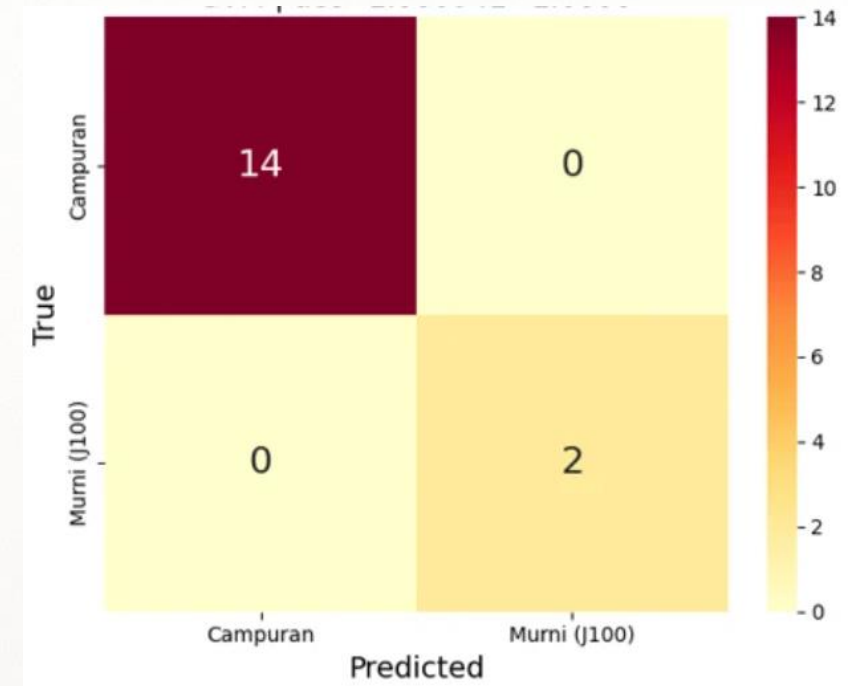
**Model system: castor oil in palm oil
8 levels (100%–30%) · 10 repeats/level ·
n = 80**

- **1450 and 1650 nm are sensitive to compositional change**
 - Absorbance increases with castor oil content.
 - 1400–1450 nm: O–H overtone of ricinoleic acid.
 - 1650–1670 nm: C–H overtone and cis unsaturation.
- **The 1720 nm channel gives the strongest absorbance, but with greater variation**
 - 1720 nm: strongest absorbance from C–H overtones in triglycerides.
 - Larger error bars at some compositions indicate lower reproducibility.

Four NIR channels for classification and quantification



Principal component scores of the four working channels.

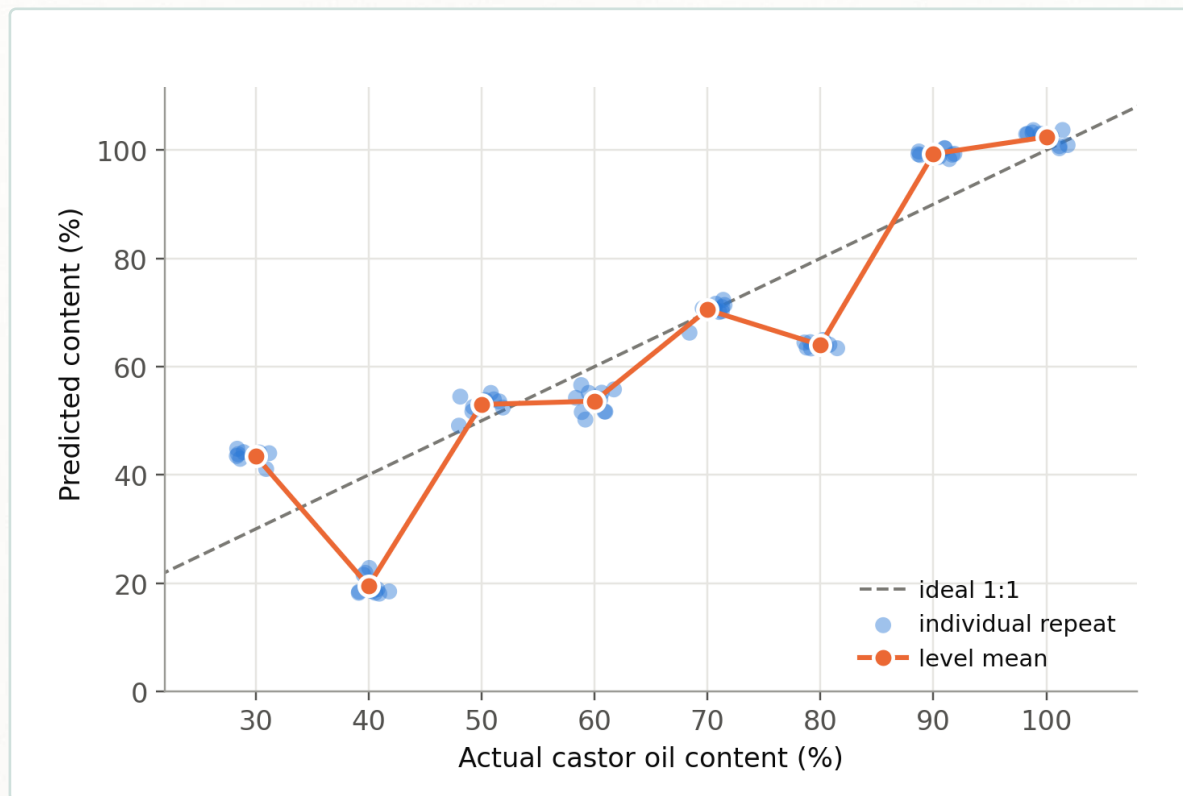


100% Classification accuracy

● The result is still preliminary

This dataset uses only one source each of castor oil and palm oil. Adding samples with a wider range of sources and characteristics is needed to make the model more robust.

Four NIR channels for classification and quantification



Castor oil content predicted using Partial Least Squares

- **The result is still preliminary**

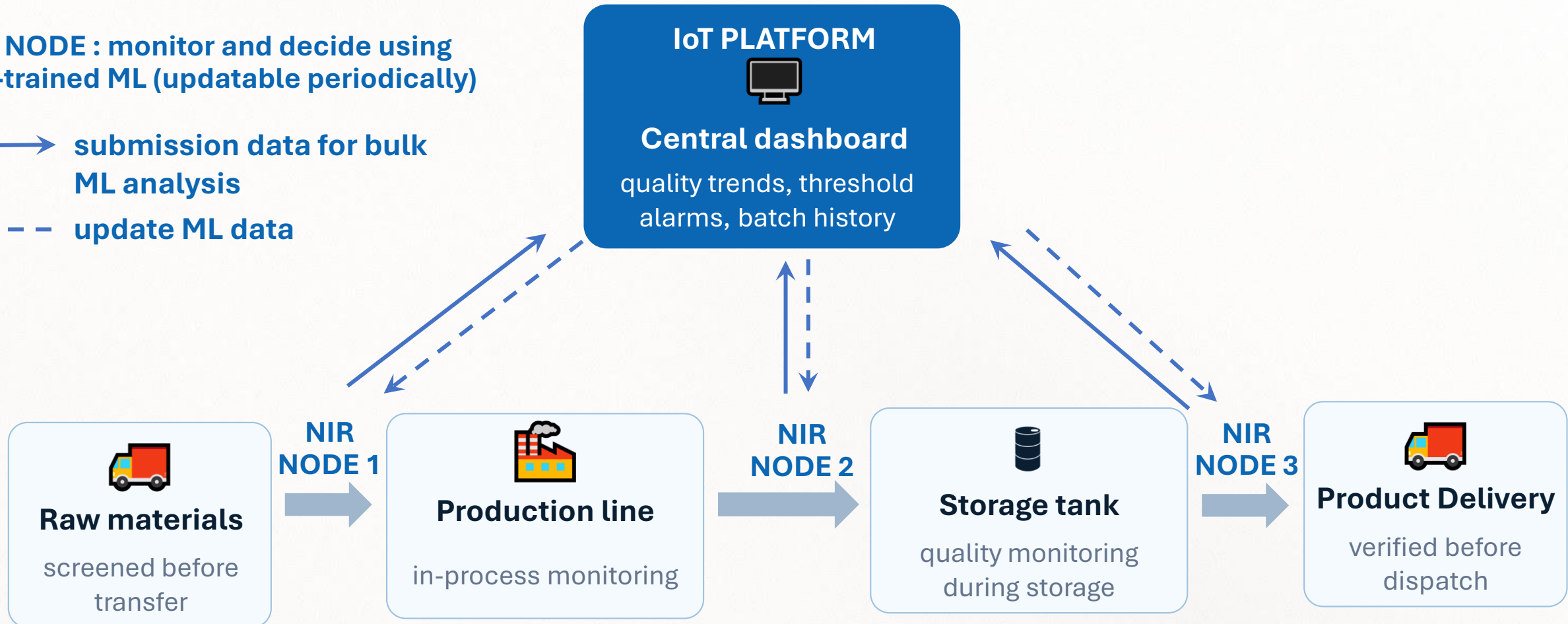
11.3% RMSE is still affected by the limited number and density of calibration points. Adding more closely spaced concentration levels and more repeats is expected to lower the prediction error.

Outlook: an IoT-based network of NIR nodes for screening and early warning

NIR NODE : monitor and decide using pre-trained ML (updatable periodically)

→ submission data for bulk ML analysis

← - - - update ML data



Thank you

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